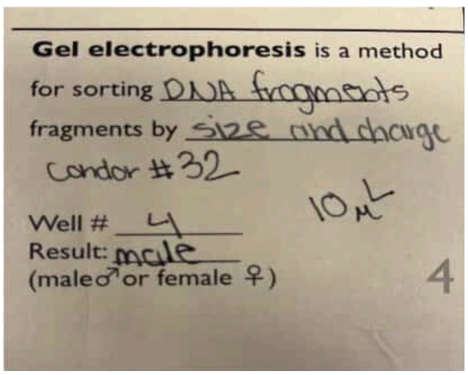
Condor Post Lab Analysis

Question: What is the gender of your condor chick?

Claim: The gender of our condor chick is a male

Evidence: 1. **Well 4**



Gel electrophoresis is a method for sorting DNA fragments fragments by size and charge

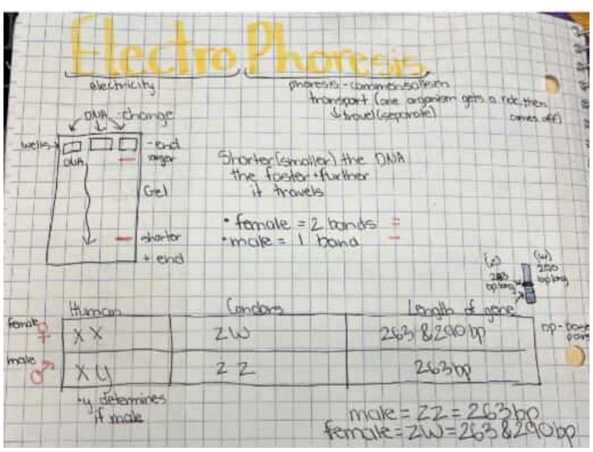
Condor #32

Well # 4 10 μL

Result: male

(male ♂ or female ♀) 4

Reasoning: The evidence supports my claim because the electrophoresis gel shows only one band of DNA, which is about 263 bp meaning it's a male (ZZ) condor chick. If it was a female (ZW) there would be one band above the other, indicating another band that's 290 bp long. Since there is only one band it means my condor chick is a male.



Analysis:

1. What are the chromosomes for a female condor?
   The chromosomes for a female condor are ZW.
2. For a male condor?
   The chromosomes for a male condor are ZZ.
3. Can you tell the gender of condors apart by looking at them?
   No you can't tell the gender of them by looking at them because they're monomorphic.
4. About how many bases long is the gene we are looking at on the z chromosome?
   We are looking at about 263 bases long on the z chromosome.
5. About how many bases long is the gene we are looking at on the w chromosome?
   We are looking at about 290 bases long on the w chromosome.
6. Which chromosome band should travel further down the gel? Why?
   The Z chromosome should travel further than the W because Z is shorter (263 bp) than W (290 bp) and shorter DNA fragments travel faster and further.